Wave Optics

Single Correct Type Questions

1. The ratio of intensities at two points P and Q on the screen in a Young's double slit experiment where phase difference between two wave of same amplitude are $\pi/3$ and $\pi/2$, respectively are **[10 April, 2023 (Shift-II)]**

(*a*) 1 : 3 (*b*) 3 : 1 (*c*) 3 : 2 (*d*) 2 : 3

2. In a Young's double slits experiment, the ratio of amplitude of light coming from slits is 2 : 1. The ratio of the maximum to minimum intensity in the interference pattern is : **[13 April, 2023 (Shift-II)]**

(*a*) 9 : 4 (*b*) 9 : 1
(*c*) 2 : 1 (*d*) 25 : 9

3. In young's double slit experiment performed using a monochromatic light of wavelength λ, when a glass plate ($\mu = 1.5$) of thickness $x\lambda$ is introduced in the path of the one of the interfering beams, the intensity at the position where the central maximum occurred previously remains unchanged. The value of x will be: **[28 June, 2022 (Shift-II)]**

(*a*) 3 (*b*) 2 (*c*) 1.5 (*d*) 0.5

4. In Young's double slit arrangement, slits are separated by a gap of 0.5 mm and the screen is placed at a distance of 0.5 m from them. The distance between the first and the third bright fringe formed when the slits are illuminated by a monochromatic light of 5890 Å is **[18 March, 2021 (Shift-I)]**

(*a*) 1178×10^{-9} m (*b*) 1178×10^{-6} m
(*c*) 1178×10^{-12} m (*d*) 5890×10^{-7} m

5. In the Young's double slit experiment, the distance between the slits varies in time as $d(t) = d_0 + a_0 \sin\omega t$; where d_0, ω and a_0 are constants. The difference between the largest fringe width and the smallest fringe width obtained over time is given as: **[25 July, 2021 (Shift-I)]**

(*a*) $\dfrac{\lambda D}{d_0 + a_0}$ (*b*) $\dfrac{\lambda D}{d_0^2} a_0$

(*c*) $\dfrac{2\lambda D a_0}{(d_0^2 - a_0^2)}$ (*d*) $\dfrac{2\lambda D(d_0)}{(d_0^2 - a_0^2)}$

6. In a Young's double slit experiment, the width of the one of the slit is three times the other slit. The amplitude of the light coming from a slit is proportional to the slit-width. Find the ratio of the maximum to the minimum intensity in the interference pattern. **[24 Feb, 2021 (Shift-I)]**

(*a*) 1:4 (*b*) 4:1 (*c*) 3:1 (*d*) 2:1

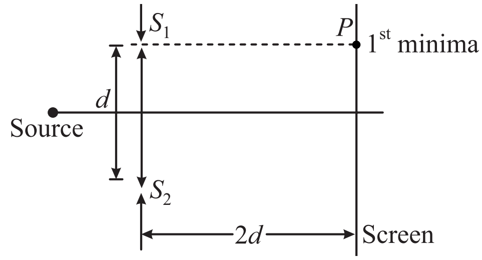
7. In a Young's double slit experiment, light of 500 nm is used to produce an interference pattern. When the distance between the slits is 0.05 mm, the angular width (in degree) of the fringes formed on the distance screen is close to: **[3 Sep, 2020 (Shift-I)]**

(*a*) 0.17° (*b*) 1.7° (*c*) 0.57° (*d*) 0.07°

8. In a Young's double slit experiment, 16 fringes are observed in a certain segment of the screen when light of a wavelength 700 nm is used. If the wavelength of light is changed to 400 nm, the number of fringes observed in the same segment of the screen would be **[2 Sep, 2020 (Shift-II)]**

(*a*) 28 (*b*) 24 (*c*) 30 (*d*) 18

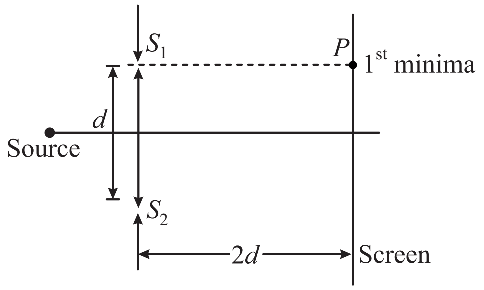
9. Consider a Young's double slit experiment as shown in figure. What should be the slit separation d in terms of wavelength λ such that the first minima occurs directly in front of the slit (S_1)? **[10 Jan, 2019 (Shift-II)]**



(*a*) $\dfrac{\lambda}{2(\sqrt{5}-2)}$ (*b*) $\dfrac{\lambda}{(\sqrt{5}-2)}$

(*c*) $\dfrac{\lambda}{2(5-\sqrt{2})}$ (*d*) $\dfrac{\lambda}{(5-\sqrt{2})}$

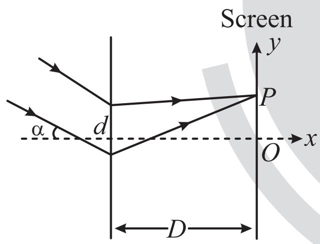
10. Consider a Young's double slit experiment as shown in figure. What should be the slit separation d in terms of wavelength λ such that the first minima occurs directly in front of the slit (S_1)? **[10 Jan, 2019 (Shift-II)]**



(*a*) $\dfrac{\lambda}{2(\sqrt{5}-2)}$ (*b*) $\dfrac{\lambda}{(\sqrt{5}-2)}$

(*c*) $\dfrac{\lambda}{2(5-\sqrt{2})}$ (*d*) $\dfrac{\lambda}{(5-\sqrt{2})}$

11. In a Young's double slit experiment, the slit separation d is 0.3 mm and the screen distance D is 1m. A parallel beam of light of wavelength 600nm is incident on the slits at angle α as shown in figure. On the screen, the point O is equidistant from the slits and distance PO is 11.0 m. Which of th following statements) is/are correct? **[JEE Adv, 2019]**



(*a*) For $\alpha = \dfrac{0.36}{\pi}$ degree, there will be destructive interference at point O.

(*b*) Fringe spacing depends on α.

(*c*) For $\alpha = \dfrac{0.36}{\pi}$ degree, there will be destructive interference at point P.

(*d*) For $\alpha = 0$, there will be constructive interference at point P.

12. In Young's double slit experiment, the fringe width is 12 mm. If the entire arrangement is placed in water of refractive index $\dfrac{4}{3}$, then the fringe width becomes (in mm): **[26 July, 2022 (Shift-I)]**

(*a*) 16 (*b*) 9

(*c*) 48 (*d*) 12

13. In Young's double slit experiment, if the source of light changes from orange to blue then: **[27 July, 2021 (Shift-I)]**

(*a*) The distance between consecutive fringes will decrease.

(*b*) The distance between consecutive fringes will increase.

(*c*) The central bright fringe will become a dark fringe.

(*d*) The intensity of the minima will increase.

14. 'n' polarizing sheets are arranged such that each makes an angle 45° with the preceeding sheet. An unpolarized light of intensity I is incident into this arrangement. The output intensity is found to be $\dfrac{I}{64}$. The value of n will be: **[1 Feb, 2023 (Shift-I)]**

(*a*) 3 (*b*) 6 (*c*) 5 (*d*) 4

15. Two polaroids A and B are placed in such a way that the pass-axis of polaroids are perpendicular to each other. Now, another polaroid C is placed between A and B bisecting the angle between them. If intensity of unpolarised light is I_0 then intensity of transmitted light after passing through polaroid B will be: **[31 Jan, 2023 (Shift-I)]**

(*a*) $\dfrac{I_0}{4}$ (*b*) $\dfrac{I_0}{2}$ (*c*) $\dfrac{I_0}{8}$ (*d*) Zero

16. An unpolarised light beam of intensity $2I_0$ is passed through a polaroid P and then through another polaroid Q which is oriented in such a way that its passing axis makes an angle of 30° relative to that of P. The intensity of the emergent light is **[29 July, 2022 (Shift-II)]**

(*a*) $\dfrac{I_0}{4}$ (*b*) $\dfrac{I_0}{2}$

(*c*) $\dfrac{3I_0}{4}$ (*d*) $\dfrac{3I_0}{2}$

17. A polarizer analyser set is adjusted such that the intensity of light coming out of the analyser is just 10% of the original intensity. Assuming that the polarizer - analyser set does not absorb any light, the angle by which the analyser need to be rotated further to reduce the output intensity to be zero is **[7 Jan, 2020 (Shift-I)]**

(*a*) 45° (*b*) 90°

(*c*) 71.6° (*d*) 18.4°

18. A system of three polarizers P_1, P_2, P_3, is set up such that the pass axis of P_3 is crossed with respect to that of P_1. The pass axis of P_2 is inclined at 60º to the pass axis of P_3. When a beam of unpolarized light of intensity I_0 is incident on P_1, the intensity of light transmitted by the three polarizers is I. The ratio (I_0/I) equals (nearly)

[12 April, 2019 (Shift-II)]

(*a*) 16.00 (*b*) 1.80

(*c*) 5.33 (*d*) 10.67

Integer Type Questions

19. In a Young's double slit experiment, the intensities at two points, for the path difference $\frac{\lambda}{4}$ and $\frac{\lambda}{3}$ (λ being the wavelength of light used) are I_1 and I_2 respectively. If I_0 denotes the intensity produced by each one of the individual slits, then $\frac{I_1+I_2}{I_0}$ =______.

[30 Jan, 2023 (Shift-II)]

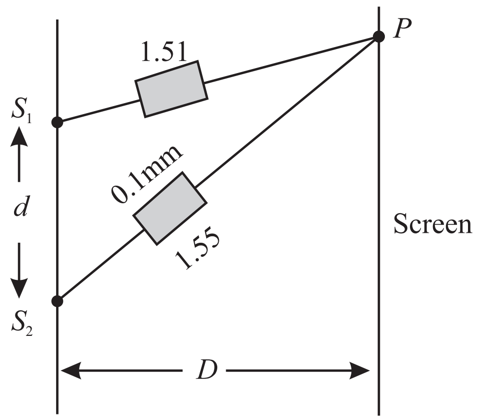
20. In a Young's double slit experiment, the slits are separated by 0.3 mm and the screen is 1.5 m away from the plane of slits. Distance between fourth bright fringes on both sides of central bright is 2.4 cm. The frequency of light used is _____ $\times 10^{14}$ Hz. [31 Aug, 2021 (Shift-II)]

21. In a Young's double's slit experiment 15 fringes are observed on a small portion of the screen when light of wavelength 500 nm is used. Ten fringes are observed on the same section of the screen when another light source of wavelength λ is used. Then the value of λ is (in nm) __________.

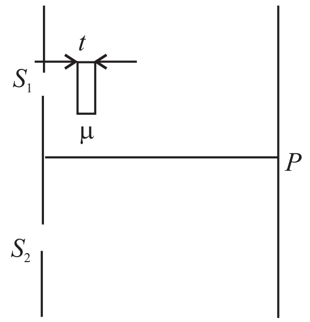
[9 Jan, 2020 (Shift-II)]

22. In Young's double slit experiment, two slits S_1 and S_2 are 'd' distance apart and the separation from slits to screen is D (as shown in figure). Now if two transparent slabs of equal thickness 0.1 mm but refractive index 1.51 and 1.55 are introduced in the path of beam ($\lambda = 4000$Å) from S_1 and S_2 respectively. The central bright fringe spot will shift by _______ number of fringes.

[30 Jan, 2023 (Shift-I)]



23. As shown in the figure, in Young's double slit experiment, a thin plate of thickness $t = 10$ μm and refractive index $\mu = 1.2$ is inserted infront of slit S_1. The experiment is conducted in air ($\mu = 1$) and uses a monochromatic light of wavelength $\lambda = 500$ nm. Due to the insertion of the plate, central maxima is shifted by a distance of $X\beta_0$. β_0 is the fringe-width before the insertion of the plate. The value of the x is _______. **[1 Feb, 2023 (Shift-II)]**



24. White light is passed through a double slit and interference is observed on a screen 1.5 m away. The separation between the slits is 0.3 mm. The first violet and red fringes are formed 2.0 mm and 3.5 mm away from the central white fringes. The difference in wavelengths of red and voilet light is ________nm. **[26 Aug, 2021 (Shift-I)]**

25. A Young's double-slit experiment is performed using monochromatic light of wavelength λ. The intensity of light at a point on the screen,where the path difference is λ, is K units. The intensity of light at a point where the path difference is $\frac{\lambda}{6}$ is given by $\frac{nK}{12}$, where n is an integer. The value of n is _________. **[6 Sep, 2020 (Shift-II)]**

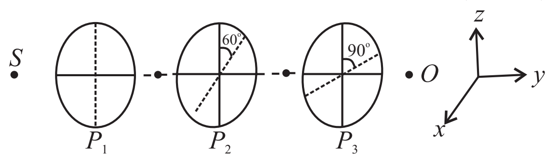
26. An unpolarised light is incident on the boundary between two dielectric media, whose dielectric constants are 2.8 (medium -1) and 6.8 (medium 2), respectively. To satisfy the condition, so that the reflected and refracted rays are perpendicular to each other, the angle of incidence should be $\tan^{-1}\left(1+\frac{10}{\theta}\right)^{\frac{1}{2}}$ the value of θ is _______.
(Given for dielectric media, $\mu_r = 1$)

[29 Jan, 2023 (Shift-II)]

27. As shown in the figures, three identical polaroids P_1, P_2 and P_3 are placed one after another. The pass axis of P_2 and P_3 are inclined at angle of 60º and 90º with respect to the axis of P_1. The source S has an intensity of $256\frac{W}{m^2}$.

The intensity of light at point O is _________ $\frac{W}{m^2}$.

[29 Jan, 2023 (Shift-I)]



28. An unpolarized light beam is incident on the polarizer of a polarization experiment and the intensity of light beam emerging from the analyzer is measured as 100 Lumens. Now, if the analyzer is rotated around the horizontal axis (direction of light) by 30^{o} in clockwise direction, the intensity of emerging light will be______ Lumens.

[24 Feb, 2021 (Shift-I)]

ANSWER KEY

1. (*c*)	**2.** (*b*)	**3.** (*b*)	**4.** (*b*)	**5.** (*c*)	**6.** (*b*)	**7.** (*c*)	**8.** (*a*)	**9.** (*a*)	**10.** (*a*)
11. (*c*)	**12.** (*b*)	**13.** (*a*)	**14.** (*b*)	**15.** (*c*)	**16.** (*c*)	**17.** (*d*)	**18.** (*d*)	**19.** [3]	**20.** [5]
21. [750]	**22.** [10]	**23.** [4]	**24.** [300]	**25.** [09]	**26.** [7]	**27.** [24]	**28.** [75]		

EXPLANATIONS

1. (*c*) $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

$= I_0 + I_0 + 2I_0\cos\phi = 2I_0(1-\cos\phi)$

For $\phi = \frac{\pi}{3} \Rightarrow I_p = 3I_0$

For $\phi = \frac{\pi}{2} \Rightarrow I_Q = 2I_0$

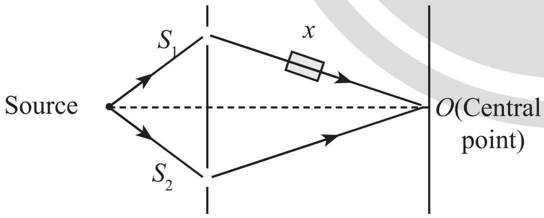
$\frac{I_P}{I_Q} = \frac{3I_0}{2I_0} = \frac{3}{2}$

2. (*b*) From the question, $\frac{A_1}{A_2} = \frac{2}{1}$

$I \propto (\text{Amplitude})^2$

$\frac{I_{\max}}{I_{\min}} = \left(\frac{A_1 + A_2}{A_1 - A_2}\right)^2 = \frac{9}{1} = 9:1$

3. (*b*)



Path difference at O = $(\mu - 1)t$.

If the intensity at O remains unchanged.

$\Rightarrow (\mu - 1)t = n\lambda$

$(1.5 - 1)\, x\lambda = n\lambda$

$\Rightarrow x = 2n$

For $n = 1, x = 2$

4. (*b*) $y_n = \frac{n\lambda D}{d}$

$\therefore 2\beta = \frac{2\lambda D}{d}$

$= 2 \times 5890 \times 10^{-10} \times \frac{0.5}{(0.5\times10^{-3})} = 1178 \times 10^{-6}$ m

5. (*c*) $\beta = \frac{\lambda D}{d} \frac{\lambda D}{d_0 + a_0 \sin\omega t}$

For the largest fringe width, $\sin\omega t = -1$

For the smallest fringe width, $\sin\omega t = +1$

$\therefore$ Difference between the largest to smallest fringe width = $\beta_1 - \beta_2$.

$\beta_1 - \beta_2 = \frac{\lambda D}{d_0 - a_0} - \frac{\lambda D}{d_0 + a_0}$

$= \lambda D\left[\frac{(d_0 + a_0) - (d_0 - a_0)}{d_0^2 - a_0^2}\right] = \frac{2\lambda D a_0}{d_0^2 - a_0^2}$

6. (*b*) Given, Amplitude, A $\propto$ slit width. We know that,

Intencity, $I \propto (A)^2$

Here, $A_1 = 3A_2$

$\frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}}\right)^2$

$= \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2} = \frac{\left(\frac{A_1}{A_2} + 1\right)^2}{\left(\frac{A_1}{A_2} - 1\right)^2} = \frac{(3+1)^2}{(3-1)^2} = 4.$

7. (*c*) $\theta = \frac{\lambda}{d} = \frac{500\times10^{-9}}{5\times10^{-5}}$

$= 0.01 \text{ rad} = 0.57°$

8. (*a*) $N_1 \lambda_1 = N_2 \lambda_2$

$16 \times 700 = N_2 \times 400 \quad \Rightarrow N_2 = 28$

9. (*a*) $\sqrt{(2d)^2 + (d)^2} - 2d = \frac{\lambda}{2}$

$\Rightarrow (\sqrt{5} - 2)\, d = \frac{\lambda}{2} \Rightarrow d = \frac{\lambda}{2(\sqrt{5}-2)}$

10. (*a*) $\sqrt{(2d)^2 + (d)^2} - 2d = \frac{\lambda}{2}$

$\Rightarrow (\sqrt{5} - 2)\, d = \frac{\lambda}{2} \Rightarrow d = \frac{\lambda}{2(\sqrt{5}-2)}$

11. (*c*) (1) $\Delta x = d\sin\alpha = d\,\alpha$ (as α is very small)

For $\alpha = \frac{0.36}{180} = (2\times10^{-3})$rad

$$\frac{\Delta x}{\lambda} = \frac{(3\times10^{-4})(2\times10^{-3})}{6\times10^{-7}} = 1 \qquad \left[\because \phi = \frac{2\pi}{\lambda}(\Delta x)\right]$$

so constructive interference

(2) $\beta = \frac{D\lambda}{d}$

(3) $\Delta x_p = d\alpha + \frac{dy}{D}$

$= 3\times10^{-4}(2\times10^{-3} + 11\times10^{-3})$

$= 39\times10^{-7}$

$\frac{\Delta x_p}{\lambda} = \frac{39\times10^{-7}}{6\times10^{-7}} = 6.5$ so destructive

(4) $\Delta x_p = \frac{dy}{D} = (3\times10^{-4})\times11\times10^{-3}$

$= 33\times10^{-7}$

$\frac{\Delta x_p}{\lambda} = \frac{33\times10^{-7}}{6\times10^{-7}} = 5.5 \Rightarrow$ destructive

12. (*b*) $\beta = 12\text{mm} = 12\times10^{-3}$ m

Formula of fringe width

$\beta = \frac{\lambda D}{d}$

Wavelength with respect to medium

$\lambda' = \frac{\lambda}{\mu}$

Fringe width in medium,

$\beta' = \frac{\lambda' D}{d} = \frac{\lambda D}{\mu d}$

$\beta' = \frac{\beta}{\mu} = \frac{12\times10^{-3}}{\frac{4}{3}} = 9\times10^{-3}\text{m}$

$\beta' = 9$ mm

13. (*a*) $\beta = \frac{\lambda D}{d}$

From formula, fringe width is directly proportional to wavelength and wavelength of orange light is greater than the blue light therefore,

So, $\beta_{\text{Orange}} > \beta_{\text{Blue}}$

Hence, distance between consecutive fringes will decrease.

14. (*b*) Intensity of light after passing through the first sheet

$I_1 = \frac{I}{2}$

Intensity of light after passing through the second sheet

$I_2 = I_1\cos^2(45°) = \frac{I}{4}$

After passing through n^{th} sheet

$I_n = \frac{I}{2^n} = \frac{I}{64}$

$\Rightarrow n = 6$

15. (*c*) Intensity of light after passing through the polariser

$I_A = \frac{I_o}{2}$

Intensity of light after passing through the polariserc

$I_C = \frac{I_o}{2}\cos^2 45 = \frac{I_o}{4}$

$\Rightarrow I_B = I_C\cos^2 45 = \frac{I_o}{8}$

16. (*c*) $I = I_0\cos^2 30°$

$= I_0\left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{4}I_0$

17. (*d*) $I = I_0\cos^2\theta$

$\frac{I_0}{10} = I_0\cos^2\theta$

$\cos\theta = \frac{1}{\sqrt{10}} = 0.31$ which is 0.707

So $\theta > 45°$ and $90 - \theta < 45°$ so only one option i.e. 18.4º

Angel rotated should be = 90º – 71.6º = 18.4º

18. (*d*) Since unpolarised light falls on $P_1 \Rightarrow$ intensity of light transmitted from $P_1(I_1) = \frac{I_0}{2}$

$\therefore$ Intensity of light transmitted form P_2

$I_2 = \frac{I_0}{2}\cos^2 30° = \frac{3I_0}{8}$

Intensity of light transmitted from P_3 is

$I = I_2\cos(60°)^2$

$\Rightarrow I = \frac{3I_o}{8}\times\frac{1}{4}$

$\Rightarrow \frac{I_o}{I} = \frac{32}{3} = 10.67$

19. [3] $\because I = 4I_0\cos^2\left(\frac{\Delta\phi}{2}\right)$ and path difference

$= \frac{\lambda}{2\pi}\times$ phase diff.

For path difference $\frac{\lambda}{4}$, phase difference $\phi = \frac{\pi}{2}$

$\Rightarrow I_1 = 4I_0\cos^2\left(\frac{\pi}{4}\right) = 2I_0$

For path difference $\frac{\lambda}{3}$, path difference $\phi = \frac{2\pi}{3}$

$$\Rightarrow I_2 = 4I_0 \cos^2\left(\frac{\pi}{3}\right) = I_0$$

$$\Rightarrow \frac{I_1 + I_2}{I_0} = 3$$

20. [5] Position of 4th bright fringe from central maxima $= \frac{4\lambda D}{d}$

As distance between 4th bright fringe on both sides of central bright fringe is $x = \frac{8\lambda D}{d}$

So, $2.4 \times 10^{-2} = \frac{8\lambda(1.5)}{0.3\times 10^{-3}} \Rightarrow \lambda = 6\times 10^{-7}$ m

Hence, $f = \frac{c}{\lambda} = \frac{3\times 10^8}{6\times 10^{-7}} = 5\times 10^{14}$ Hz

21. [750] We will use the formula, $\frac{n_1\lambda_1 D}{d} = \frac{n_2\lambda_2 D}{d}$

[D is the distance from the screen to the slits and d is the separation between the slits.]

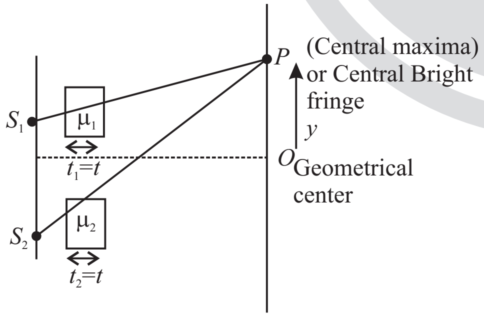
$$15\times 500\times\frac{D}{d} = 10\times\lambda_2\times\frac{D}{d}$$

$\lambda_2 = 15 \times 50$ nm

$\lambda_2 = 750$ nm

Hence, value of λ is 750.

22. [10]



Path difference at P be Δx

$$\Delta x = (\mu_2 - \mu_1)t \;= (1.55 - 1.51)0.1\text{mm}$$

$= 0.04 \times 10^{-4}$

$$\Delta x = 4\times 10^{-6} = 4\mu m$$

$$y = \frac{\Delta x D}{d} = 4\times 10^{-6}\frac{D}{d}$$

{y is the distance of central maxima from geometric center}

fringe width $\underset{(\beta)}{}$ $= \frac{\lambda D}{d} = 4\times 10^{-6} m\frac{D}{d} = 4\mu m\frac{D}{d}$

$\therefore$ Central bright fringe spot will shift by 'x'

Number of shift $= \frac{y}{\beta}$

$$= \frac{4\times 10^{-6} D/d}{4\times 10^{-7} D/d} = 10$$

23. [4] Fringe shift $= \frac{t(\mu - 1)}{\lambda}\beta_o$

$$= \frac{10\times 10^{-6}(1.2-1)}{5\times 10^{-7}}\beta_o = 4\beta_o$$

24. [300] For first fringe of Red wavelength

$$\frac{dy_1}{D} = 1\times\lambda_1 \qquad [n = 1]$$

For first fringe of violet wavelength

$$\frac{dy_2}{D} = 1\times\lambda_2 \qquad [n = 1]$$

$$\frac{d}{D}(y_1 - y_2) = \lambda_1 - \lambda_2$$

$$y_1 - y_2 = \frac{D}{d}(\lambda_1 - \lambda_2)$$

$$= \frac{0.3\times 10^{-3}}{1.5}(3.5-2)\times 10^{-3}$$

$= 300 \times 10^{-9} = 300$ nm

25. [09] $\phi = \frac{2\pi}{\lambda}\times\frac{\lambda}{6} = \frac{\pi}{3}$

$$I = k\cos^2\left(\frac{\phi}{2}\right) = \frac{3K}{4}$$

$n = 9$

26. [7]

$$\Rightarrow \tan i = \frac{\mu_2}{\mu_1} = \sqrt{\frac{6.8}{2.8}}$$

$$\tan i = \left(\frac{2.8+4}{2.8}\right)^{1/2} \Rightarrow i = \tan^{-1}\left(1+\frac{10}{7}\right)^{1/2}$$

$$\Rightarrow \theta = 7$$

27. [24] After passing through the first polaroid, intensity of the light is halved $= \dfrac{256}{2}$

Polaroid P_2 and P_3 will make intensity $\cos^2(60^o)$ and $\cos^2(30^o)$ times respectively.

$\Rightarrow$Intensity of light point O

$$= \frac{256}{2} \times \frac{1}{4} \times \left(\frac{\sqrt{3}}{2}\right)^2 = 24$$

28. [75] Using Malus' law,

$$I = I_0 \cos^2\theta = 100 \times \cos^2(30^o) = 75 \text{ Lumens}$$